

Adam Bawatneh

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Website | GitHub | LinkedIn

Research Interests

Large language and multimodal models, social reasoning, benchmark construction and evaluation, computer vision, responsible AI.

Education

University of Central Florida

Ph.D. in Computer Science

GPA: 4.00/4.00

Orlando, Florida

Fall 2025 – Present

University of Central Florida

B.S. in Computer Science

Orlando, Florida

Spring 2025

Selected Publication

Adam Bawatneh, Sagar Sapkota, Amrit Singh Bedi, Santu Karmaker, and Mubarak Shah. “OmniToM: Benchmarking Theory of Mind in LLMs via Explicit Belief Modeling.” Accepted to *NeurIPS 2026*, Evaluations & Datasets Track.

[arXiv:2605.26322](#) | [Project page](#) | [Benchmark](#)

Research Experience and Projects

OmniToM: LLM Social Reasoning Evaluation Benchmark

Oct 2025 – Present

Research project, University of Central Florida

- Co-developed a benchmark from 895 stories with 22,343 structured belief propositions and 156,401 labels, using a human-validated annotation pipeline developed with 11 annotators and about 1.1K person-hours of annotation work.
- Coordinated annotation quality control and validation, and designed structured prompts for belief extraction and seven-dimension belief labeling.
- Evaluated API and open-weight models across Azure OpenAI, Anthropic, Google Vertex AI, and university and NSF ACCESS HPC resources; built parallel and batch evaluation workflows for model comparison.

BIAS-101: Demographic Bias in Video Action Recognition

Jan 2025 – Apr 2025

Research project, University of Central Florida

- Built a video collection and filtering pipeline using platform APIs, automated quality checks, vision-language models, and multimodal annotation to audit demographic representation in UCF-101.
- Processed more than 85,000 candidate videos and added more than 2,000 curated clips for dataset rebalancing; assessed demographic patterns with statistical tests and a dominance-ratio measure.

DETECT: Deception Tracking Through Eye Cues Technology

Aug 2024 – Apr 2025

Faculty-advised, 6-member team project

- Led development of a real-time gaze and facial-signal tracking system using MediaPipe FaceMesh, with reported gaze-tracking accuracy above 80% in the project evaluation.
- Built and deployed the full-stack application using Astro, Go, PostgreSQL, Docker, and DigitalOcean.

CLEAR: Video Denoising Evaluation

Research project, University of Central Florida

Compared traditional and AI-based denoisers on more than 13,000 videos, using Python and OpenCV to generate noise conditions and PSNR, SSIM, and VIF to evaluate output quality.

Teaching Experience

Undergraduate Teaching Assistant, University of Central Florida

May 2023 – Spring 2025

Computer Science courses in C, Python, and Java

Supported lab instruction and student mentoring in course sections serving more than 1,500 students.

Technical Skills

Programming: Python, C/C++, Java, JavaScript, Go, SQL

AI and evaluation: LLM/VLM evaluation, benchmark design, prompt engineering, multimodal annotation, computer vision

Platforms and APIs: Azure OpenAI, Anthropic API, Google Vertex AI, Hugging Face, Azure, Google Cloud, HPC

Tools: Git/GitHub, Linux, Docker, PostgreSQL, REST APIs, LaTeX, Jira